CS228 Logic for Computer Science
Exercise Problem Set 3

Part 1

1. Prove (using Natural Deduction) or disprove (using argument you think is convincing enough) whether the following sequents are valid:
   (a) $A \vdash A \to B$
   (b) $A \vdash B \to A$
   (c) $A \wedge B \vdash A \to B$
   (d) $\vdash (A \vee B) \to A$
   (e) $\vdash (A \vee (B \wedge A)) \to A$

2. Translate the following reasoning into a sequent. If the sequent is valid, prove it using the rules of natural deduction. If the sequent is not valid, provide a counter example.

   If I press the button, the window opens.
   The window is open.
   Therefore, I pressed the button.

3. Convert the following formulas to equivalent formulas in Conjunctive Normal Form (CNF) and Disjunctive Normal Form (DNF). Show your intermediate steps.
   (a) $\neg(\neg(p \wedge q) \to \neg r)$
   (b) $\neg(p \to (q \wedge r))$
   (c) $(p \to q) \to r$
   (d) $\neg(\neg p \vee (q \to \neg r))$

4. Convert the following formulae into CNF form:

$$\begin{aligned}
\varphi_1 &:= x_1 \to (x_2 \vee x_3) \\
\varphi_2 &:= x_2 \to (\neg x_3 \vee \neg x_4) \\
\varphi_3 &:= x_3 \to (\neg x_2 \vee x_4) \\
\varphi_4 &:= x_4 \to \neg(x_1 \vee x_2) \\
\varphi_5 &:= \neg x_4 \to \neg(\neg x_3 \vee x_3)
\end{aligned}$$

   Suppose $\varphi := \bigwedge_{i=1}^{5} \varphi_i$, then write φ in CNF form.

Part 2

5. Apply Tseitin's encoding to the following formulas to convert them to CNF. How many new variables did you need?

   (a) $\Phi := \neg\left[(a \vee \neg b) \wedge \neg((a \vee \neg c) \vee d)\right]$

   (b) $\Psi := \neg(P \vee (\neg Q \wedge R)) \rightarrow (\neg P \wedge (Q \vee \neg R))$

   We saw in class that Tseitin's transformation preserves satisfiability. Does it also preserve validity? Prove or disprove.

6. In the class, we have discussed about two normal forms, namely, CNF and DNF. In this question, we introduce another one called **Algebraic Normal Form (ANF)**. Informally, ANFs are expressions involving $\oplus$ (xor) and $\wedge$ (conjunction) connectives. For example, $y = (x_1 \wedge x_2) \oplus x_3$ is in ANF. More formally, a well formed formula ϕ over propositional variables $x_1, x_2, \ldots, x_n$ in ANF form is written as:

$$\phi(x_1, x_2, \ldots, x_n) = c_0 \oplus \bigoplus_{1 \leq i \leq n} (c_i \wedge x_i) \oplus \bigoplus_{1 \leq i,j \leq n} (c_{ij} \wedge x_i \wedge x_j) \oplus \cdots \oplus (c_{1\ldots n} \wedge x_1 \wedge x_2 \cdots \wedge x_n),$$

   where each constant literal $c_t \in \{\bot, \top\}$. It can be proven that every wff of n variables can be uniquely represented in this form. As an example, the formula $x_1 \rightarrow x_2$ can be written as $\top \oplus x_1 \oplus (x_1 \wedge x_2)$; you can check that the truth tables for both of these formulae are the same.

   Now, convert the following Boolean function into its equivalent ANF form:

$$\phi(x_0, x_1, x_2) = (\neg x_0 \wedge \neg x_1 \wedge \neg x_2) \vee (\neg x_0 \wedge \neg x_1 \wedge x_2) \vee (x_0 \wedge \neg x_1 \wedge x_2)$$

7. Recall the distributive law $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$, which lets us convert any CNF formula into an equivalent DNF formula by repeatedly distributing $\wedge$ over $\vee$ (and symmetrically for DNF to CNF). Recall also that the **length** of a CNF/DNF formula is the total number of literals occurring in it (e.g., $(p \vee q) \wedge (\neg p \vee r \vee q)$ has length 5).

   (a) Convert $(x_1 \vee y_1) \wedge (x_2 \vee y_2)$ into DNF using the distributive law.

   (b) Give an example of a family of CNF formulae $\{\psi_n : n \in \mathbb{N}\}$ over $2n$ variables such that distributing ψ_n out fully produces a DNF formula with 2^n terms. (It suffices to exhibit the direct expansion; you need not argue that no smaller equivalent DNF exists.)

   (c) Give an example of a family of DNF formulae $\{\phi_n : n \in \mathbb{N}\}$ over $2n$ variables and of length $2n$, such that *every* equivalent CNF formula over the same variables — not just one obtained by naive distribution — has exponential length (in n).

8. Consider the parity function, PARITY : $\{0,1\}^n \rightarrow \{0,1\}$, where PARITY evaluates to 1 iff an odd number of inputs is 1. In all of the CNFs below, we assume that each clause contains any variable at most once, i.e. no clause contains expressions of the form $p \wedge \neg p$, or $p \vee \neg p$. Furthermore, all clauses are assumed to be distinct.

   (a) Prove that any CNF representation of PARITY must have n literals (from distinct variables) in every clause.

   (b) Prove that any CNF representation of PARITY must have at least 2^{n-1} clauses.